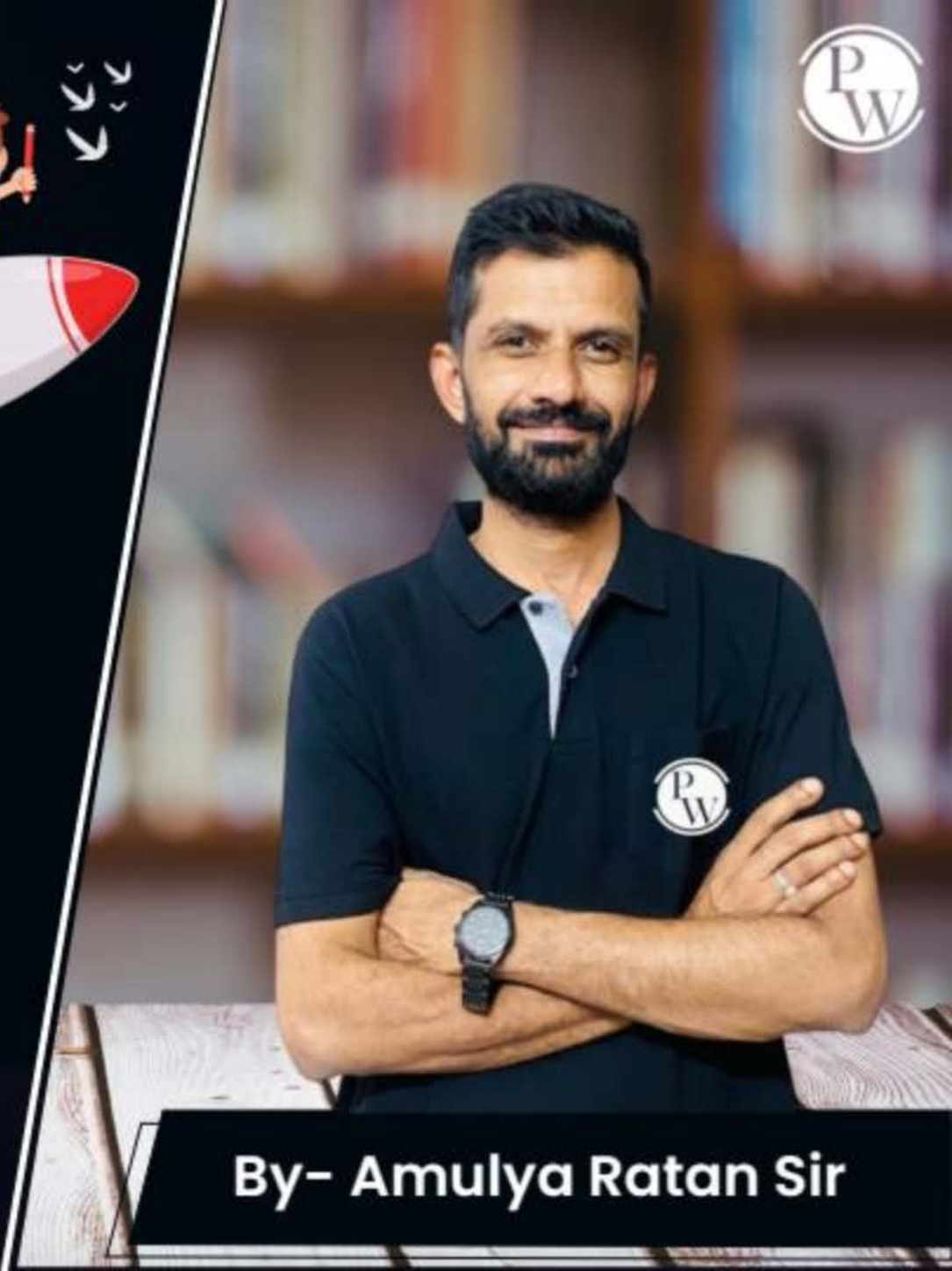
# GATE ALL BRANCHES



## General Aptitude

## QUANTITATIVE APTITUDE

Lecture No.- 01



By- Amulya Ratan Sir

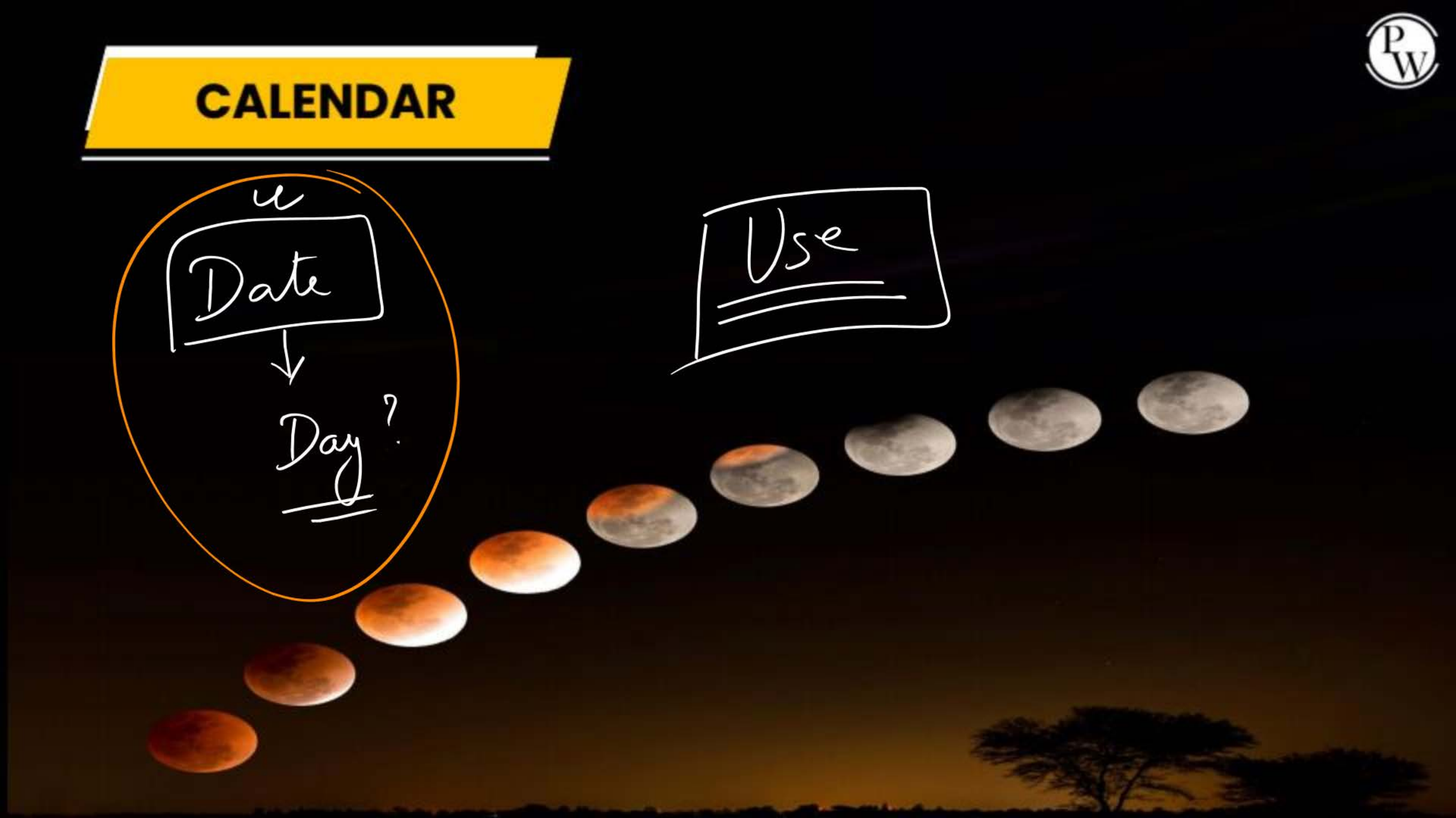
# Topics to be Covered



- **Topic-1** Basic of English Calendar
- **Topic-2** Importance of ODD Days
- **Topic-3** Calendar Repetition
- **Topic-4** Questionnaire



# CALENDAR



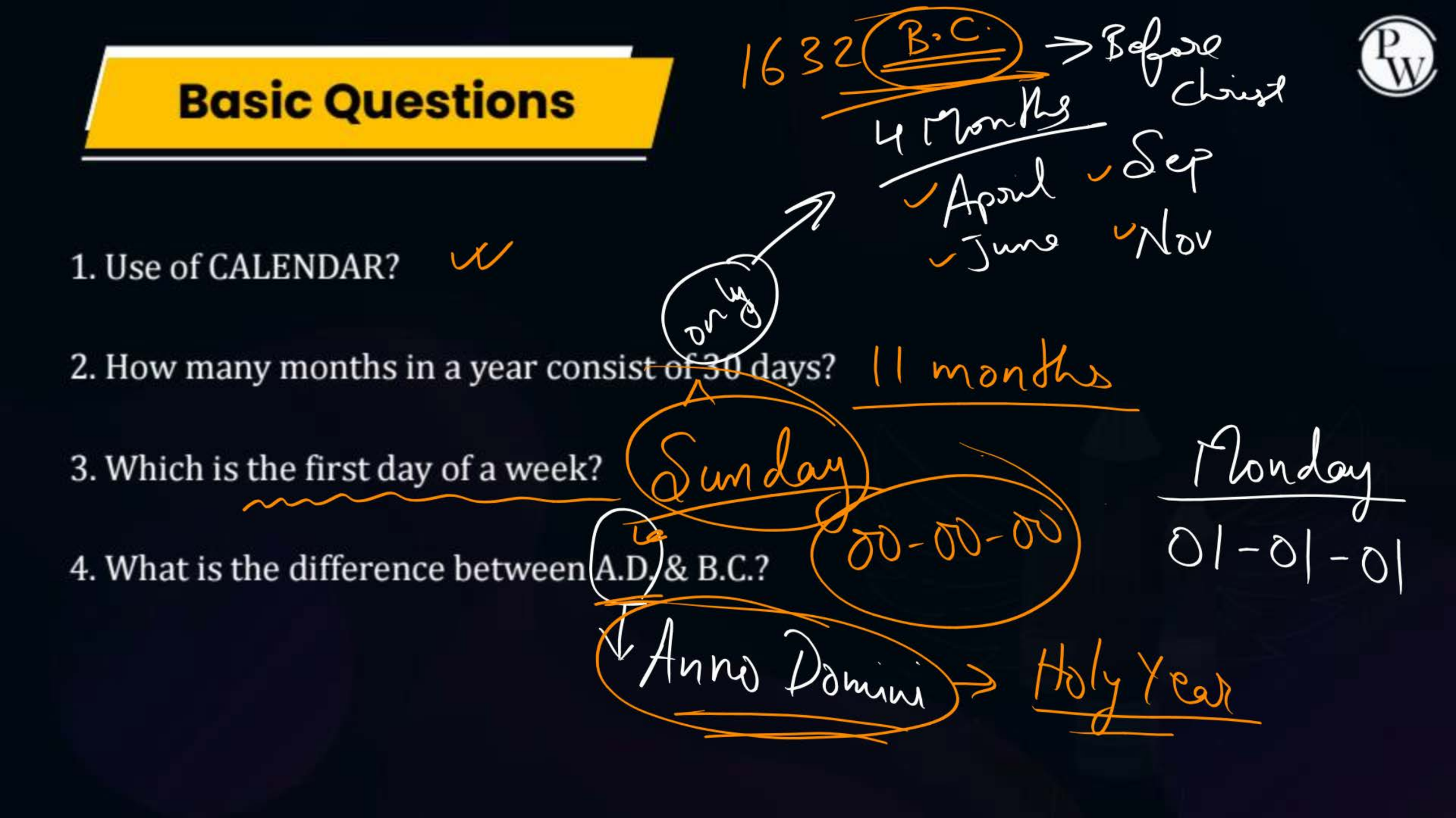
# Basic Questions

1. Use of CALENDAR?

2. How many months in a year consist of 30 days?

3. Which is the first day of a week?

4. What is the difference between A.D. & B.C.?



# LEAP YEAR Occurrence

4

400

1862 ✗

Minimum → 1 year

1712 ✓

Maximum → 7 ~~4~~ 8 years

1796 → 29th Feb

~~1800~~ 1804 → 29th Feb

# To be Noted:



365 days 5 hrs 42 min -- secs

$365\frac{1}{4}$ days

OR

365 days 6 hrs

✓If a year is divisible by 4 then it's a leap year else normal. e.g. 1652, 1212, 1496, 1708, etc. are leap years whereas 1714, 1446, 2006 etc. are normal years.

✓If a century is divisible by 400 then it's a leap year else normal. e.g. 1600, 1200, 2000, 800 etc. are leap years whereas 1000, 1500, 2100 etc. are normal years.

6 hrs
6 hrs
6 hrs
6 hrs

24 hrs or 1 day

400

L.Y. ← 4

1 → 365
2 → 365
3 → 365
4 → 365 + 1 = 366

# ODD DAYS

? ODD DAYS

As today is 16th November, 2023. The day is Thursday. +3

✓What would be the day on 26th November, 2023? Sunday

10 days

7

3 days

04 08 12 16 20 24 28 32 ... 96

# ODD DAYS

÷7

Remainder

7) 30 (4 weeks
28
---
2 → odd days

- ✓ All those number of days which can't be kept in a group of a week.

OR

- ✓ When given number of days is divided by 7, the remainder is called as odd days.

60

45

4 Y

4

400

# Normal Year & Leap Year

~~52 weeks~~

N.Y. ⇒ 365 ⇒ 1 odd days

L.Y. ⇒ 366 ⇒ 2 odd days

# First Century

ODD DAYS → 0-6

7) days ( 

R = ?

100 years → 5 odd days

L.Y. 24 (x2) 48

N.Y. 76 (x1) 76

12 = 6 + 6

04
08
12
16
20
24
28
32

24 → 96

# Odd Days in Centuries

CALENDAR

Date → Day

1300 900 500 X 100 → 5 — 1700

1400 1000 600 X 200 ⇒ 3 — (10) 1800

1500 1100 700 X 300 ⇒ 1 — (15) 1900

1600 1200 800 X 400 ⇒ 0 — 2000

500 → 5

6+1

Odd day

# Questionnaire:

1 odd day → Monday

25th Oct, 2004

3 + 5 = 8

0 - Sun
1 - Mon
2 - Tue
3 - Wed
4 - Thu
5 - Fri
6 - Sat

2003 ⇒ 3 odd days

2000 ⇒ 0

3 yrs ⇒ 3 odd days

| | | | |
|---|---|---|---|
| (31) J - 3 | M - 3 | S - 2 | |
| (29) F - 1 | J - 2 | | |
| (31) M - 3 | J - 3 | 0 ⇒ 4 | |
| (30) A - 2 | A - 3 | | |

5 odd days

# Questionnaire:

$5+6=11$

10th March, 2005 → 4 odd days

2004 ⇒ 5

J - 3

F - 0

M - 3

6

2000 ⇒ 0

Thursday

4 year → 5 ✓

L.Y | N.Y

1 | 3

(2) (3)

3rd June, 1947

1. Tuesday
2. Friday
3. Sunday
4. Wednesday

2 + 0 = (2)

1946 → 2

1900 → 1

J - 3
F - 0
M - 3
A - 2
M - 3
J - 3

46 → 1

L.Y  N.Y.

11  35

22  35

1  0

Tuesday (XII)

04

44

# Questionnaire:

4 + 0 = 4 odd days

#Q. Guru Nanak was born on 15th April, 1469. What was the week day? Thursday

1468 ⇒ 4 odd days

1400 ⇒ 3

68 → 1

J - 3

F - 0

M - 3

A - 1

Q

L.Y N.Y

17

34

6

51

1

+ 2 = 8

× 17

68

**Fact:**

Diary

1st pattern

2011

Date

2nd pattern

Day 2.




## April 15, 1469: Day of the Week

*April 15, 1469* was the 105th day of the year 1469 in the Gregorian calendar. There were 260 days remaining until the end of the year. The day of the week was *Thursday*.

The day of the week for April 15, 1469 under the old Julian calendar was Saturday. Did you notice the difference with the Gregorian calendar?

# CALENDAR REPETITION

2017 - 1

2018 - 1

2019 - 1

2020 - 2

2021 - 1

2011 → 1

2012 → 2

2013 → 1

2014 → 1

2015 → 1

2016 → 2

2022

# Questionnaire:

2011

2005 → 1

2006 → 1

2007 → 1

2008 → 2

2009 → 1

2010 → 1

[MCQ]

#Q. Mahatma Gandhi was born on 2nd October 1869. What was the week Day?

Assignment

A Monday

B Thursday

C Tuesday

D Saturday

[MCQ]

#Q. The day of the March 16th of any year is the same day of the week as the corresponding date in which month of the same year?

Assignment

A July

B November

C April

D September

[MCQ]

#Q. If January 1st, 1992 was a Wednesday. What day of the week was January 1st, 2003?

Assignment

A Sunday

B Wednesday

C Thursday

D Friday



# 2 mins Summary

**Topic** Calendar

## ODD DAYS

1. Date ⇔ Day?
2. Calendar Repeats

Cont...

# THANK - YOU